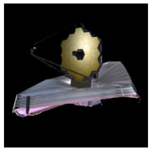

9016 - Galaxy Formation at The Redshift Frontier: Ultra-Deep NIRSpec Observations of z>13 galaxies

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				
	1	GOODSS z14	NIRSpec MultiObject Spectroscopy	(3) GS_z14_catalog

ABSTRACT

Over the last two years, JWST has discovered an unexpected population of bright galaxies at z>10. The first look at the spectra of this population (GNz11, GHZ2) was surprising, revealing extreme lines that point to gas properties and ionizing sources not seen at lower redshift. These extreme spectra are thought to be powered by strong bursts of star formation or AGN, both of which may explain the large luminosities. The newly-discovered galaxy GS-z14-0 at z=14.2 presents a departure from these systems. It is extremely bright, but it is mostly featureless in the NIRSpec discovery spectrum, with only a single CIII] detection. How GS-z14-0 can appear so luminous at z=14.2 without exhibiting the extreme spectra seen in similarly bright z>10 galaxies is not known. It is plausible it is a strong burst (similar to GNz11 and GHZ2), but with a slightly less extreme population of ionizing agents. Unfortunately current data do not place strong constraints on the specific star formation rate or metallicity of GS-z14-

0. Here we propose to obtain the spectroscopy necessary to characterize the ionizing sources and gas conditions in GS-z14-0. With 45.3 hours of R=1000 G235M spectroscopy with the NIRSpec MSA, we will place the first robust constraints on a wide range of physical properties. These observations will provide first window on galaxy formation at z=14, while also providing one of our only spectroscopic benchmarks against which theories of why z>10 galaxies are so luminous can be compared. We will simultaneously observe 39 galaxies at z>5, including two additional z>13 sources, providing the deepest view to-date of the gas and ionizing sources at very high redshift.

OBSERVING DESCRIPTION

We propose ultra-deep NIRSpec MSA observations targeting GS-z14-0 using the G235M grating.

The MSA pointing center and PA are set to simultaneously observe GS-z14-0, GS-z14-1, and GS-z13-LA1. Remaining available space on the MSA is filled with spectroscopically confirmed-galaxies at z>4 identified from shallow wide-field grism spectroscopy or NIRSpec/MSA PRISM observations. We will use the G235M/F170LP grating/filter combination to target the rest-frame UV emission from the primary high-redshift sources. Meeting our science goals requires reaching a 3sigma limit on the equivalent width of emission lines in the rest-fram UV of 1.3AA requiring a limiting line flux of 6x10^-20 erg/s/cm^2. This will require a total exposure time of 50 hours with NIRSpec with G235M. Existing spectroscopy comprises 4.7 hours in the same grating with NIRSpec, thus we require an additional 45.3 hours of exposure time. Remaining space on the MSA is filled with galaxies at z>4 identified from shallow wide-field grism spectroscopy in addition to bright z>4 LBGs.

Proposal 9016 - Targets - Galaxy Formation at The Redshift Frontier: Ultra-Deep NIRSpec Observations of z>13 galaxies

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	GS_z14_catalog	RA: 03 32 34.4300 (53.1434583d) Dec: -27 47 47.01 (-27.79639d) Equinox: J2000		

Comments:
Description=[]

Observation

Proposal 9016, Observation 1: GOODSS z14

Tue Mar 11 23:05:46 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec MultiObject Spectroscopy

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	GS_z14_catalog	RA: 03 32 34.4300 (53.1434583d) Dec: -27 47 47.01 (-27.79639d) Equinox: J2000		

Comments:
Description=[]

Acquisition

#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
2		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	

Template

TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold
MSATA	false	No	MSA Center	GS_z14_catalog (8941 sources)		jwst-nirspec-g235m	1.5

Reference Stars

Proposal 9016 - Observation 1 - Galaxy Formation at The Redshift Frontier: Ultra-Deep NIRSpec Observations of z>13 galaxies

Spectral Elements

#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
1	1 (G235M/F170LP)	c1	3 Shutter Slitlet	53.09070666666666 Degrees -27.87039722222221 Degrees	190.02466891007273			3	12	27310.402
2	1 (G235M/F170LP)	c1	3 Shutter Slitlet	53.09070666666666 Degrees -27.87039722222221 Degrees	190.02466891007273			3	12	27310.402
3	1 (G235M/F170LP)	c1	3 Shutter Slitlet	53.09070666666666 Degrees -27.87039722222221 Degrees	190.02466891007273			3	12	27310.402
4	1 (G235M/F170LP)	c1	3 Shutter Slitlet	53.09070666666666 Degrees -27.87039722222221 Degrees	190.02466891007273			3	12	27310.402
5	1 (G235M/F170LP)	c1	3 Shutter Slitlet	53.09070666666666 Degrees -27.87039722222221 Degrees	190.02466891007273			3	12	27310.402
6	1 (G235M/F170LP)	c1	3 Shutter Slitlet	53.09070666666666 Degrees -27.87039722222221 Degrees	190.02466891007273			3	12	27310.402

Special Requirements

Group Visits within 53.0 Days
Visits Same PA
MSA Planned Aperture PA 190.0000 to 190.0000 Degrees (V3 51.4254303 to 51.4254303)